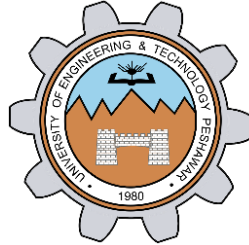


Lab # 03

Bubble, Selection, Insertion Sort



Spring 2026

Data Structures and Algorithms Lab

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Task 01: Implementation of Bubble Sort in Arrays:

```
lab_03_sorting_algorithms > bubble_sort_T_01.cpp > bubbleSort(int [], int)
You, 2 hours ago | 1 author (You)
1 #include <iostream>
2 using namespace std;
3 void bubbleSort(int arr[], int size)
4 {
5     for (int i = 0; i < size - 1; i++)
6     {
7         for (int j = i + 1; j < size; j++)
8         {
9             int temp;
10            if (arr[i] > arr[j])
11            {
12                temp = arr[i];
13                arr[i] = arr[j];
14                arr[j] = temp;
15            }
16        }
17    }
18    for (int i = 0; i < size; i++)
19    {
20        cout << arr[i] << " ";
21    }
22 }
You, 3 hours ago • bubble sort implemented
23 int main()
24 {
25     int arr[] = {3, 5, 1, 2, 6, 4};
26     int size = sizeof(arr) / sizeof(arr[0]);
27     bubbleSort(arr, size);
28 }
```

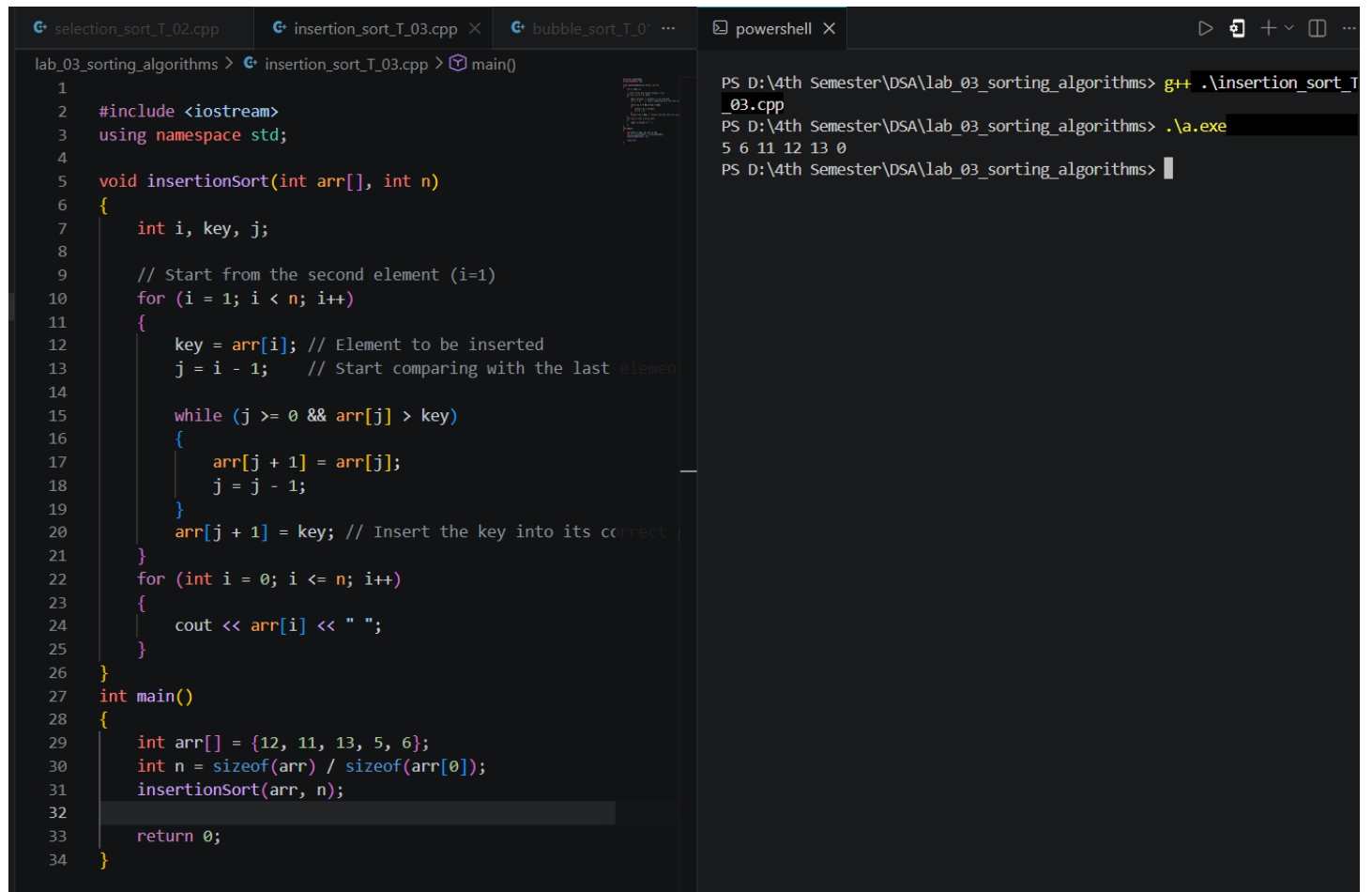
```
2 files changed, 34 insertions(+)
create mode 100644 lab_03_sorting_algorithms/insertion_sort_T_03.cpp
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> git push -u origin mai
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 98.60 KiB | 7.04 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mohsinkhan117/CPP_DSA.git
1ea16d2..ff1aa26 main -> main
branch 'main' set up to track 'origin/main'.
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> g++ .\bubble_sort_T_01
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> .\a.exe
1 2 3 4 5 6
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> 
```

Task 02: Selection sort implementation:

```
lab_03_sorting_algorithms > selection_sort_T_02.cpp > main()
1 #include <iostream>
2 using namespace std;
3
4 void selectionSort(int arr[], int size)
5 {
6     for (int i = 0; i < size - 1; i++)
7     {
8         int min = i;
9
10        for (int j = i + 1; j < size; j++)
11        {
12            if (arr[j] < arr[min])
13            {
14                min = j;
15            }
16        }
17
18        if (min != i)
19        {
20            swap(arr[i], arr[min]);
21        }
22    }
23 }
24
25 int main()
26 {
27     int arr[] = {3, 5, 1, 2, 6, 4};
28     int size = sizeof(arr) / sizeof(arr[0]);
29
30     selectionSort(arr, size);
31
32     for (int i = 0; i < size; i++)
33     {
34         cout << arr[i] << " ";
35     }
36
37     return 0;
38 }
```

```
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> g++ .\selection_sort
_02.cpp
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> .\a.exe
1 2 3 4 5 6
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> 
```

Task 03: Implementation of Insertion Sort:



The screenshot shows a C++ IDE with three tabs: 'selection_sort_T_02.cpp', 'insertion_sort_T_03.cpp', and 'bubble_sort_T_01...'. The 'insertion_sort_T_03.cpp' tab is active, displaying the following code:

```
1
2 #include <iostream>
3 using namespace std;
4
5 void insertionSort(int arr[], int n)
6 {
7     int i, key, j;
8
9     // Start from the second element (i=1)
10    for (i = 1; i < n; i++)
11    {
12        key = arr[i]; // Element to be inserted
13        j = i - 1;    // Start comparing with the last element
14
15        while (j >= 0 && arr[j] > key)
16        {
17            arr[j + 1] = arr[j];
18            j = j - 1;
19        }
20        arr[j + 1] = key; // Insert the key into its correct position
21    }
22    for (int i = 0; i <= n; i++)
23    {
24        cout << arr[i] << " ";
25    }
26 }
27 int main()
28 {
29     int arr[] = {12, 11, 13, 5, 6};
30     int n = sizeof(arr) / sizeof(arr[0]);
31     insertionSort(arr, n);
32
33     return 0;
34 }
```

To the right of the code editor is a PowerShell terminal window. It shows the execution of the program:

```
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> g++ .\insertion_sort_T_03.cpp
PS D:\4th Semester\DSA\lab_03_sorting_algorithms> .\a.exe
5 6 11 12 13 0
PS D:\4th Semester\DSA\lab_03_sorting_algorithms>
```

Comparison :

Feature	Bubble	Selection	Insertion
Best Case	$O(n)$	$O(n^2)$	$O(n)$
Average Case	$O(n^2)$	$O(n^2)$	$O(n^2)$
Swaps/Moves	Many	Few swaps	Fewer operations
Stable	Yes	No	Yes
Adaptive	Yes	No	Yes
Practical Performance	Poor	Fair	Good

RUBRICS (LABS/ PROJECTS):

Criteria & Point Assigned	Outstanding 4	Acceptable 3	Considerable 2	Below Expectations 1	Score
Valuing related to work, including responsibility, and professionalism.	Demonstrates excellent responsibility and professionalism in task management. Shows strong commitment to quality work, meeting the deadline, and comprehensive, well-organized effective documentation following all guidelines with exemplary attention to detail.	Demonstrates some responsibility and professionalism. Displays moderate commitment to quality work, meets the deadline and shows sufficient thoroughness in documentation, adhering to guidelines, with some room for enhancement.	Demonstrates limited effort in responsibility and professionalism. Misses the deadline with notable inconsistency in quality work and shows insufficient thoroughness or clarity in documentation, necessitating improvements.	Demonstrates a lack of responsibility and professionalism. Misses the deadline and lacks quality work. Documentation is incomplete, disorganized, or unclear.	
Ability to apply techniques, methods, and procedures effectively.	Executes lab tasks with exceptional skill to design/develop solution and shows proficiency in lab techniques and procedures.	Executes lab tasks with adequate skill to design/develop solution and shows proficiency in lab techniques and procedures.	Executes lab tasks with minimal skill to design/develop solution and shows moderate proficiency in lab techniques and procedures.	Executes lab tasks with limited skill to design/develop solution and lacks proficiency in lab techniques and procedures.	

Demonstrate understanding and relevant Knowledge.	Demonstrates exceptional understanding of concepts with clear explanation of investigation and analysis of the objectives and outcomes.	Demonstrates a good understanding of concepts with some explanations of investigation and analysis that align with the lab's objectives and outcomes.	Demonstrates a basic understanding of key concepts but lacks clarity or depth in explaining the investigation and analysis, with some inconsistencies.	Demonstrates limited understanding of concepts with unclear explanations of investigation and analysis that do not align with the lab's objectives and outcomes.	
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